
System Requirements Specification

for

SafeOvertake_Thymio

Version 1.0

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**(a modified version of IEEE Software Requirements
Specification Template, K.E. Wieggers, 1999)**

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Revision History

Name	Date	Reason for Changes	Version
Muhammad Zayyan Shafiq Rehan Abu Hashir	18 May 2026	Initial SRS draft prepared from project assignment, SRS template, and current Safe_Overtake_Thymio implementation.	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for SafeOvertake_Thymio, a Thymio Playground simulator prototype for a safe overtake scenario on a two-lane undivided road. The SRS covers the simulated system-of-systems composed of three Thymio vehicles, the road/environment model, the line-following and proximity-based coordination behavior, and the requirements that connect the implementation to the course project target.

The document is intended to be editable and suitable for submission or further refinement. It focuses on the requirements specification document required by the course project. It also records traceability to the implemented Aseba and Playground artifacts supplied in the project ZIP.

1.2 Document Conventions

Requirement identifiers use the form REQ-**<VIEWPOINT>**-**<number>**, for example REQ-ARCH-01.

Priority values are High, Medium, and Low. High requirements are essential to satisfy the safe overtake target.

The words shall and must indicate mandatory requirements. Should indicates a recommended requirement. May indicates an optional capability.

The system-of-systems terminology follows the course concepts: SoS, Constituent System (CS), Relied Upon Interface (RUI), Relied Upon Message Interface (RUMI), and Relied Upon Physical Interface (RUPI).

Unless explicitly stated otherwise, requirements apply to the simulator prototype and not to real road vehicles.

1.3 Intended Audience and Reading Suggestions

This document is intended for the project evaluator, course instructor, student developer, and any reader who needs to understand, test, or modify the SafeOvertake_Thymio prototype.

Course evaluator: read Sections 1, 3, 4, 5, and 6 to understand scope, use cases, requirements, traceability, and validation.

Developer/student: read Sections 2, 4, and Appendix A to connect requirements to the Thymio/Aseba implementation.

Tester: read Sections 4, 6, and Appendix B to execute validation scenarios and check expected results.

Modeling reviewer: read Sections 2.1, 4.1, 4.3, and 4.6 to understand the SoS viewpoints used for the conceptual model.

1.4 Product Scope

SafeOvertake_Thymio demonstrates a cyber-physical system-of-systems in which autonomous constituent vehicles coordinate through sensing and the shared physical road environment to achieve the higher-level goal of avoiding collisions during an overtaking maneuver.

Vehicle A is the slow vehicle in the original/right lane.

Vehicle B is the faster vehicle initially behind A and is responsible for initiating the overtaking maneuver.

Vehicle C is the opposite-direction vehicle in the left/opposite lane.

The road is a two-lane undivided road represented by a Playground world and SVG ground texture.

The prototype shall show that B can detect A, overtake when safe, react to C if C appears during the overtake, and merge back into the original lane without colliding.

The following are out of scope for this SRS: real vehicle deployment, real road-law certification, wireless vehicle-to-vehicle communication, multi-lane intersections, pedestrian behavior, GPS time synchronization, and formal safety certification.

1.5 References

ID	Reference
REF-01	Course project slides: Project Description A-Lab Introduction, Project “safe overtake”.
REF-02	SRS template supplied by the course: srs_template - V2.0.doc, modified IEEE Software Requirements Specification Template.
REF-03	Course slides: Requirements Engineering - 2.0.
REF-04	Course slides: Interfaces in Evolving Cyber-Physical Systems of Systems.
REF-05	Course slides: Basics on Cyber-Physical SoSs.
REF-06	Implementation artifact: Safe_Overtake_Thymio.zip, containing A.aesl, B.aesl, C.aesl, calibrate.aesl, safe_overtake.playground, and straight_road.svg.

2. Overall Description

2.1 Product Perspective

SafeOvertake_Thymio is a new, self-contained simulator prototype created for the course project. It is not a production traffic-control system. It is a controlled demonstrator for requirements engineering, SoS modeling, and Thymio-based implementation.

The system is best described as a directed prototype SoS: the vehicles are implemented and configured by the student, while each simulated vehicle behaves as an autonomous Constituent System during execution. Coordination is performed mainly through physical/stigmergic channels: each vehicle observes the road markings and nearby vehicles through sensors and affects the shared environment through movement, LEDs, and sound.

Figure 1. SafeOvertake_Thymio system context and SoS boundary.

2.2 Product Functions

Load a Thymio Playground world with a two-lane undivided road and three vehicles A, B, and C.

Keep vehicles aligned with lane markings using proportional line following based on ground sensors.

Allow A to move slower than B, creating an overtaking need.

Allow B to detect A using proximity sensors when B becomes sufficiently close.

Allow B to execute a staged overtake: pull left, align in the opposite lane, pass A, merge back right, and resume line following.

Allow B to detect C during critical overtake phases using front/left proximity sensors.

Allow B to clear the opposite lane instead of stopping in C's lane when C creates danger.

Stop a vehicle when a wall or strong frontal obstacle is detected.

Provide visible LED and sound feedback for turning, braking, and completed overtake states.

Provide a calibration helper for inspecting ground/proximity sensor values before tuning thresholds.

2.3 User Classes and Characteristics

User Class	Characteristics	Main Needs
Student Developer	Has basic knowledge of Thymio Suite, Aseba Studio, Playground, and course SoS concepts.	Modify thresholds, speeds, timers, and initial vehicle positions; demonstrate the scenario.
Course Instructor / Evaluator	Reviews requirements, model consistency, implementation behavior, and oral explanation.	Verify that the project target is satisfied and that artifacts are complete.
Tester / Peer Reviewer	Can run the simulation and observe vehicle behavior but may not modify code.	Execute test cases and report whether expected behaviors occur.
Future Maintainer	May extend the prototype with message interfaces, additional vehicles, or more road layouts.	Understand requirement IDs, assumptions, implementation mapping, and extension points.

2.4 Operating Environment

Element	Specification
Simulator	Thymio Suite / Aseba Playground capable of loading .playground worlds and .aesi programs.
Robots	Three simulated Thymio II robots named Vehicle_A, Vehicle_B, and Vehicle_C.
World file	safe_overtake.playground with world size 150 x 400 and ground texture straight_road.svg.
Road model	Straight two-lane undivided road, black lane surface, white outer boundaries, and dashed white center line.
Programs	A.aesi for Vehicle A, B.aesi for Vehicle B, C.aesi for Vehicle C, and calibrate.aesi for sensor calibration.
Host system	Desktop/laptop able to run Thymio Suite. The prototype does not require external network access.
Ports	Playground robots are configured on local ports 33334 (A), 33333 (B), and 33335 (C).

2.5 Design and Implementation Constraints

The implementation shall use Thymio/Aseba-style event handlers, variables, subroutines, proximity sensors, ground sensors, timers, LEDs, sound, and motor targets.

The prototype shall be executable in the simulator without requiring physical Thymio robots.

The road shall remain simple enough for ground-sensor line following; complex curves, traffic signs, pedestrians, and intersections are excluded.

The overtaking behavior may be implemented through timed phases rather than a fully continuous planner.

The prototype shall not rely on direct digital communication among vehicles; the current coordination shall be achieved through sensors and the shared environment.

Thresholds and timers shall be tunable because sensor values and motion timing depend on the simulator configuration.

The SRS shall follow the supplied course template and organize system requirements using SoS viewpoints.

2.6 User Documentation

No separate user manual is required. Appendix A maps each implementation artifact to its requirements. Appendix B lists current implementation parameter values. Appendix C provides a glossary of SoS terminology used in this document.

2.7 Assumptions of Use

Vehicle A starts in front of B in the original/right lane, and Vehicle B starts behind A.

Vehicle B has a higher nominal base speed than Vehicle A.

Vehicle C, if present in the scenario, travels in the opposite/left lane toward B.

Lane markings are visible to the ground sensors, and thresholdLeft/thresholdRight are calibrated for the current road texture.

The simulator executes events and timers consistently enough for the defined overtaking phases to be observable.

No human operator intervenes during the test except to start/stop the simulation or load code.

The prototype is evaluated by inspection and simulation, not by formal road safety certification.

3. System Use Cases

The major system use cases focus on the visible behavior required to demonstrate the safe overtake target. The use case diagram is intentionally high-level and does not model every internal timer step as a separate use case.

Figure 2. High-level use case diagram for SafeOvertake_Thymio.

3.1 Run Safe Overtake Simulation (U1)

Field	Description
Objective	The operator loads the Playground world and vehicle programs, starts the simulation, and observes the full scenario.
Priority	High

Actors	Student/Operator; Aseba Playground; Vehicle A; Vehicle B; Vehicle C.
Basic Flow	1. Open safe_overtake.playground in Aseba Playground. 2. Load A.aesl to Vehicle_A, B.aesl to Vehicle_B, and C.aesl to Vehicle_C. 3. Start the simulator execution. 4. Observe vehicles moving on the road according to their assigned behaviors. 5. Stop the simulation after B completes the scenario or after a test condition has been observed.
Alternative Flow(s)	If sensor values do not match the road texture, run calibrate.aesl and tune thresholds before repeating the test.
Exception Flow(s)	If the simulator does not show all vehicles or code is not attached, reload the world and verify the robot ports/names.
Includes	U2, U3, U4, U5, U6, U7 as applicable.
Preconditions	Thymio Suite/Aseba Playground is installed; project files are available.
Postconditions	The scenario is executed and visible evidence is available for evaluation.
Notes/Issues	This use case supports demonstration and validation rather than normal end-user operation.

3.2 Maintain Lane Using Ground-Sensor Line Following (U2)

Field	Description
Objective	Each active vehicle follows its lane without drifting outside the road under nominal conditions.
Priority	High
Actors	Vehicle A, Vehicle B, Vehicle C; Road/environment.
Basic Flow	1. The vehicle receives ground sensor updates. 2. The controller reads left and right ground sensor values. 3. The controller computes proportional error relative to thresholds. 4. The controller applies a dead zone for small errors. 5. The controller sets left and right motor targets to correct position.
Alternative Flow(s)	If the vehicle is intentionally turning for an overtake phase, timer-based steering temporarily overrides normal line following.
Exception Flow(s)	If a wall or strong obstacle is detected, U6 overrides line following and stops the vehicle.
Includes	U6 when frontal obstacle is detected.
Preconditions	Vehicle is on a lane with detectable markings; thresholds are calibrated.
Postconditions	Vehicle remains aligned with the lane or transitions safely to an override behavior.
Notes/Issues	The current code uses prox.ground.delta[0] and prox.ground.delta[1].

3.3 Detect Slower Vehicle A and Initiate Overtake (U3)

Field	Description
Objective	Vehicle B detects when it is sufficiently close to Vehicle A and initiates the overtake maneuver.
Priority	High

Actors	Vehicle B; Vehicle A; Road/environment.
Basic Flow	1. B follows the original/right lane behind A. 2. B reads frontal proximity sensor data. 3. When the center frontal sensor exceeds the configured collisionDistance threshold, B marks carDetected = 1. 4. B activates the left indicator. 5. B enters the first overtake phase and starts steering left.
Alternative Flow(s)	If A is not detected, B continues line following. If a wall is detected instead of a vehicle, B stops using U6.
Exception Flow(s)	If the threshold is too low, B may initiate early; if too high, B may approach A too closely. Calibration and test cases must check this.
Includes	U2 before detection; U4 after detection.
Preconditions	B is behind A and has not already completed the overtake.
Postconditions	B enters the timed overtake state sequence.
Notes/Issues	In the implementation, Vehicle B uses collisionDistance = 400 for A detection.

3.4 Perform Overtake and Merge Back (U4)

Field	Description
Objective	Vehicle B overtakes A by entering the opposite lane and then returns to the original/right lane after passing A.
Priority	High
Actors	Vehicle B; Vehicle A; Road/environment.
Basic Flow	1. B pulls left from the original lane. 2. B turns right to align in the opposite lane. 3. B drives straight in the overtake lane to pass A. 4. B turns right to merge back into the original lane. 5. B turns left to straighten inside the original lane. 6. B resumes normal forward movement and line following.
Alternative Flow(s)	If C is detected during the align/pass phases, B executes U5 before resuming. If B has not fully cleared A, timing parameters must be adjusted and the test repeated.
Exception Flow(s)	If B follows the dashed center line instead of stabilizing in the lane, lane-centering thresholds or phase timing require adjustment.
Includes	U5 when danger is detected; U2 after overtake completion.
Preconditions	U3 has detected A and phase = 1 has started.
Postconditions	B is back in the original/right lane and overtake is marked complete or reset after emergency.
Notes/Issues	B.aesl implements phases 1 to 5 for nominal overtaking and phases 6 to 8 for emergency recovery.

3.5 Detect Opposite Vehicle C and Clear Danger Lane (U5)

Field	Description
Objective	When C is detected during an overtake, B shall avoid stopping in the opposite lane and shall move right to clear C's lane.

Priority	High
Actors	Vehicle B; Vehicle C; Road/environment.
Basic Flow	1. During phase 2 or 3, B checks left/front/center proximity sensors. 2. If any monitored sensor exceeds dangerDistance, B sets dangerDetected = 1. 3. B activates the right indicator. 4. B steers right to leave the opposite lane. 5. B straightens and resumes normal behavior after the emergency sequence.
Alternative Flow(s)	If C is not detected, B continues the nominal overtake sequence. If C appears after B has returned to the original lane, B continues normal line following unless a frontal obstacle is detected.
Exception Flow(s)	If B detects A as danger after passing A, sensor selection and thresholds must avoid false danger detection from behind/right-side objects.
Includes	U2 after recovery.
Preconditions	B is in phase 2 or phase 3 of the overtake sequence.
Postconditions	B has cleared C's lane and dangerDetected/emergencyStop flags are reset after recovery.
Notes/Issues	The implementation intentionally uses left/front/center sensors for C detection to reduce confusion with A on B's right side.

3.6 Stop on Wall or Strong Frontal Obstacle (U6)

Field	Description
Objective	A vehicle stops and shows brake indication when a strong wall-like obstacle is detected in front.
Priority	Medium
Actors	Any vehicle; Road/environment.
Basic Flow	1. Vehicle computes the average of five front horizontal proximity sensors. 2. If the average exceeds stopDistance, hasDetectedWall is set to 1. 3. The vehicle stops both motors. 4. The brake LED indication is shown.
Alternative Flow(s)	If no wall is detected, normal line following continues.
Exception Flow(s)	Close vehicles may sometimes resemble walls; the threshold must be tuned to avoid unwanted stopping during the overtake.
Includes	None.
Preconditions	Vehicle code is running and proximity events are available.
Postconditions	Vehicle is stopped when a wall-like obstacle is detected.
Notes/Issues	A.aesl and C.aesl use the same wall check structure as the base line-following behavior.

3.7 Calibrate Sensors (U7)

Field	Description
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Objective	The operator inspects current ground and proximity sensor values before tuning thresholds.
Priority	Medium
Actors	Student/Operator; Thymio robot; Road/environment.
Basic Flow	1. Load calibrate.aesl on a selected robot. 2. Run the simulator with the robot over relevant road colors and obstacle positions. 3. Observe LED circle values corresponding to ground and front sensor readings. 4. Record values for thresholdLeft, thresholdRight, dangerDistance, collisionDistance, and stopDistance tuning.
Alternative Flow(s)	If values are unstable, repeat calibration at different positions on the road texture.
Exception Flow(s)	If sensor events do not fire, verify simulator connection and robot selection.
Includes	None.
Preconditions	Aseba/Playground can execute calibrate.aesl.
Postconditions	Threshold values can be adjusted based on observed sensor values.
Notes/Issues	Calibration supports repeatability but is not part of the autonomous driving behavior.

4. System Requirements Definition

Requirements are organized by SoS-inspired viewpoints. Each requirement is concise, verifiable, and traced to upper-layer project requirements and test cases in later sections.

Figure 3. Vehicle B nominal and emergency behavior sequence.

Figure 4. Initial simulation scene derived from the Playground and road texture files.

4.1 Architecture Viewpoint

Requirement ID	Priority	Requirement Statement	Verification Method
REQ-ARCH-01	High	The system shall be modeled as a Safe Overtake System-of-Systems composed of Vehicle A, Vehicle B, Vehicle C, and the shared road/environment.	Simulation observation / code inspection / test case
REQ-ARCH-02	High	Each vehicle shall be treated as a Constituent System with its own controller, sensors, actuators, and behavior logic.	Simulation observation / code inspection / test case
REQ-ARCH-03	High	The road/environment shall be treated as the Environmental Constituent System that enables lane following and stigmergic coordination.	Simulation observation / code

			inspection / test case
REQ-ARCH-04	High	Vehicle B shall be the primary overtaking CS responsible for detecting A, performing the overtake, reacting to C, and returning to the original lane.	Simulation observation / code inspection / test case
REQ-ARCH-05	Medium	The architecture shall allow A, B, and C to run independent Aseba programs so that each CS remains autonomous during simulation.	Simulation observation / code inspection / test case
REQ-ARCH-06	Medium	The prototype shall include all executable artifacts required to run the simulator: Playground world, road texture, and Aseba programs.	Simulation observation / code inspection / test case

4.2 Functional and Behavioral Viewpoint

Requirement ID	Priority	Requirement Statement	Verification Method
REQ-FUNC-01	High	Vehicles A, B, and C shall maintain lane alignment using proportional line following based on the two ground sensors.	Simulation observation / code inspection / test case
REQ-FUNC-02	High	Vehicle A shall move slower than Vehicle B in the nominal scenario.	Simulation observation / code inspection / test case
REQ-FUNC-03	High	Vehicle B shall detect Vehicle A using frontal proximity sensing when A is within the configured collisionDistance threshold.	Simulation observation / code inspection / test case
REQ-FUNC-04	High	Vehicle B shall initiate an overtake when Vehicle A is detected and no wall-stop condition is active.	Simulation observation / code inspection / test case
REQ-FUNC-05	High	Vehicle B shall execute the nominal overtake through staged movement: pull left, align in overtake lane, pass, merge right, straighten, and resume line following.	Simulation observation / code inspection / test case

REQ-FUNC-06	High	Vehicle B shall monitor for Vehicle C during the critical aligning and passing phases of the overtake.	Simulation observation / code inspection / test case
REQ-FUNC-07	High	When Vehicle C is detected during the critical overtake phases, Vehicle B shall clear the opposite lane by steering right before resuming movement.	Simulation observation / code inspection / test case
REQ-FUNC-08	High	Vehicle B shall not intentionally stop in the opposite lane when C creates a collision danger.	Simulation observation / code inspection / test case
REQ-FUNC-09	Medium	All vehicles shall stop both motors when a strong wall-like frontal obstacle is detected.	Simulation observation / code inspection / test case
REQ-FUNC-10	Medium	The system shall provide visual indication for left turn, right turn, braking, and overtake completion states.	Simulation observation / code inspection / test case
REQ-FUNC-11	Medium	The system shall provide a calibration mode to observe ground and proximity sensor values.	Simulation observation / code inspection / test case

4.3 Interface and Communication Viewpoint

Requirement ID	Priority	Requirement Statement	Verification Method
REQ-INT-01	High	The RUPI of each vehicle shall include ground sensors for observing lane markings on the physical road model.	Simulation observation / code inspection / test case
REQ-INT-02	High	The RUPI of each vehicle shall include horizontal proximity sensors for observing nearby vehicles and wall-like obstacles.	Simulation observation / code inspection / test case
REQ-INT-03	High	The RUPI of each vehicle shall include motor actuators for influencing the shared physical environment through movement.	Simulation observation / code

			inspection / test case
REQ-INT-04	Medium	The RUPI of each vehicle shall include LED and sound actuators for observable state indication during demonstration.	Simulation observation / code inspection / test case
REQ-INT-05	High	The current prototype shall not require direct vehicle-to-vehicle RUMI messages; coordination shall be achieved through the shared physical environment.	Simulation observation / code inspection / test case
REQ-INT-06	Medium	The SRS shall preserve RUMI as an extension point for future direct cyber-channel communication among vehicles.	Simulation observation / code inspection / test case
REQ-INT-07	High	Vehicle B shall use front/left/center proximity sensing for C danger detection to reduce false detection of A after or beside B.	Simulation observation / code inspection / test case

4.4 Time and Synchronization Viewpoint

Requirement ID	Priority	Requirement Statement	Verification Method
REQ-TIME-01	High	Vehicle B shall use timed phases to structure the overtaking maneuver.	Simulation observation / code inspection / test case
REQ-TIME-02	High	Vehicle B shall check for C during phase 2 and phase 3, where the collision risk with the opposite lane is highest.	Simulation observation / code inspection / test case
REQ-TIME-03	High	The emergency give-way sequence shall interrupt the nominal phase sequence when dangerDetected becomes true.	Simulation observation / code inspection / test case
REQ-TIME-04	Medium	Timer values shall be configurable so that the maneuver can be tuned for road dimensions and simulated vehicle speed.	Simulation observation / code inspection / test case

REQ-TIME-05	Medium	The simulator shall provide sufficient event timing consistency for proximity events and timer0 transitions to be observable.	Simulation observation / code inspection / test case
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4.5 Dependability, Safety, and Risk Viewpoint

Requirement ID	Priority	Requirement Statement	Verification Method
REQ-SAFE-01	High	The system shall avoid collisions between B and A during approach, overtaking, and merge-back scenarios under the defined assumptions.	Simulation observation / code inspection / test case
REQ-SAFE-02	High	The system shall avoid collisions between B and C when C appears during B's overtake under the defined assumptions.	Simulation observation / code inspection / test case
REQ-SAFE-03	High	Vehicle B shall prioritize clearing the opposite lane over completing the overtake when C is detected.	Simulation observation / code inspection / test case
REQ-SAFE-04	Medium	The system shall expose thresholds and timer constants so that unsafe early/late reactions can be corrected during calibration.	Simulation observation / code inspection / test case
REQ-SAFE-05	Medium	The system shall include a wall-stop behavior to avoid collision with fixed frontal obstacles.	Simulation observation / code inspection / test case
REQ-SAFE-06	Medium	The prototype shall support repeatable test scenarios by defining initial vehicle positions in the Playground file.	Simulation observation / code inspection / test case
REQ-SAFE-07	Low	The SRS shall include known hazards and mitigation requirements for the simulator prototype.	Simulation observation / code inspection / test case

4.6 Dynamicity, Evolution, and Emergence Viewpoint

Requirement ID	Priority	Requirement Statement	Verification Method
REQ-EVO-01	Medium	The system shall tolerate scenario dynamicity where C may or may not be present during B's overtake.	Simulation observation / code inspection / test case
REQ-EVO-02	Medium	The system shall support future evolution by keeping each vehicle behavior in a separate Aseba file.	Simulation observation / code inspection / test case
REQ-EVO-03	Medium	The system shall support future evolution by allowing the road texture and Playground positions to be changed independently from controller logic.	Simulation observation / code inspection / test case
REQ-EVO-04	Medium	The desired emergent behavior shall be safe traffic flow: B completes or aborts/recovers from the overtake without collision.	Simulation observation / code inspection / test case
REQ-EVO-05	Low	The prototype shall document negative emergence risks such as false danger detection, following the center line, or phase timing mismatch.	Simulation observation / code inspection / test case

4.7 Security and Misuse Constraints

Requirement ID	Priority	Requirement Statement	Verification Method
REQ-SEC-01	Low	The prototype shall not store credentials, personal data, or network secrets.	Simulation observation / code inspection / test case
REQ-SEC-02	Low	The prototype shall run locally and shall not require external network communication.	Simulation observation / code inspection / test case
REQ-SEC-03	Low	The documented misuse cases shall include manual modification of thresholds, wrong file loading, and wrong initial positions as configuration risks.	Simulation observation / code

			inspection / test case
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4.8 Hazard and Risk Summary

The following hazard summary supports the safety/dependability requirements and helps justify the simulator test plan.

Hazard ID	Hazard	Potential Cause	Effect	Mitigation / Related Requirements
HZ-01	B collides with A	B detects A too late; collisionDistance too low; B speed too high.	Collision or blocked overtake.	REQ-FUNC-03, REQ-FUNC-04, REQ-SAFE-01, TC-03.
HZ-02	B collides with C	C appears while B is in the opposite lane; B stops in C lane.	Head-on collision risk.	REQ-FUNC-06, REQ-FUNC-07, REQ-FUNC-08, REQ-SAFE-02, TC-05.
HZ-03	B follows dashed center line	Line following resumes while B is centered on lane marking.	Unstable lane position or wrong lane tracking.	REQ-FUNC-05, REQ-TIME-04, REQ-SAFE-04, TC-04, TC-09.
HZ-04	False danger detection	B detects A or road boundary as C due to sensor selection/thresholds.	Unnecessary emergency maneuver.	REQ-INT-07, REQ-SAFE-04, TC-05, TC-09.
HZ-05	Obstacle/wall collision	Fixed obstacle or boundary detected too late.	Vehicle impact or invalid scenario.	REQ-FUNC-09, REQ-SAFE-05, TC-07.

5. Traceability Matrix

The upper-layer requirements below are derived from the course project target and assignment. They are traced to SRS requirement groups and use cases.

Upper-Layer Req.	Upper-Layer Requirement	This SRS Requirements	Use Cases
UL-01	The scenario shall use a road with two undivided lanes.	REQ-ARCH-01, REQ-ARCH-03, REQ-INT-01	U1, U2
UL-02	Vehicles A and B shall start on the same lane, with A in front of B.	REQ-ARCH-01, REQ-SAFE-06	U1, U3
UL-03	B shall move faster than A.	REQ-FUNC-02	U2, U3
UL-04	When B is sufficiently close to A, B shall overtake A.	REQ-FUNC-03, REQ-FUNC-04, REQ-FUNC-05, REQ-TIME-01	U3, U4

UL-05	If C comes in the opposite direction, B and C shall avoid collision.	REQ-FUNC-06, REQ-FUNC-07, REQ-FUNC-08, REQ-SAFE-02, REQ-SAFE-03	U5
UL-06	The project shall include a requirements specification document using the required template.	All SRS sections; REQ-SAFE-07	U1
UL-07	The system specification should use SoS viewpoints and distinguish SoS, CS, RUMI, and RUPI requirements.	REQ-ARCH-01 to REQ-ARCH-06; REQ-INT-01 to REQ-INT-07; REQ-EVO-01 to REQ-EVO-05	U1
UL-08	The prototype shall be implemented using Thymio and simulator artifacts.	REQ-ARCH-06, REQ-FUNC-01 to REQ-FUNC-11, REQ-TIME-01 to REQ-TIME-05	U1–U7

6. Test Cases / Test Plan

Although test cases were marked as outside the minimum project assignment in the template, they are included here to make the requirements verifiable and easier to demonstrate during the oral interview.

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Results
TC-01	Load complete simulation	Open safe_overtake.playground; attach A.aesl, B.aesl, C.aesl to corresponding robots; run.	All project files available.	World starts with road texture and three vehicles.
TC-02	Verify lane following	Run A and C individually on the road; observe motion.	thresholdLeft=75; thresholdRight=77; base speeds from code.	Vehicles follow their lane without wall-stop.
TC-03	Verify B detects A	Run full scenario with B behind A.	A ahead of B; B baseSpeed=65; A baseSpeed=28.	B detects A and begins overtake before collision.
TC-04	Nominal overtake without C danger	Place C far away or remove C; run B behind A.	Default A/B positions; C not in danger path.	B pulls left, aligns, passes A, merges right, straightens, resumes.

TC-05	Emergency C detection during overtake	Place C in opposite lane so B detects it during phase 2 or 3.	dangerDistance around 300; C approaching.	B steers right to clear the opposite lane and does not stop in C's lane.
TC-06	Recovery after emergency	Continue TC-05 until emergency sequence finishes.	Same as TC-05.	B resets emergency flags and resumes normal following/retry behavior.
TC-07	Wall/obstacle stop	Place a wall-like obstacle in front of a vehicle or move vehicle toward boundary.	stopDistance=3000.	Vehicle stops both motors and shows brake LED.
TC-08	LED indications	Observe B during left turn, right turn, brake, and overtake completion states.	Default B.aesl.	Left/right LEDs, brake top LED, and all-lights completion state appear.
TC-09	Sensor calibration/tuning	Load calibrate.aesl and observe LED values over lane colors and near obstacles.	Robot on road and near target objects.	Operator can determine whether thresholds need adjustment.
TC-10	Initial positions and file mapping	Inspect safe_overtake.playground and confirm robot names/ports.	A at original lane ahead; B behind; C opposite lane.	Robot names match A/B/C code assignment and positions match scenario assumptions.

6.1 6.1 Test Cases / Test Plan Traceability Matrix

SRS Requirement(s)	Test Case(s)
REQ-ARCH-01 to REQ-ARCH-06	TC-01, TC-10
REQ-FUNC-01	TC-02
REQ-FUNC-02 to REQ-FUNC-05	TC-03, TC-04
REQ-FUNC-06 to REQ-FUNC-08	TC-05, TC-06
REQ-FUNC-09	TC-07
REQ-FUNC-10	TC-08
REQ-FUNC-11	TC-09
REQ-INT-01 to REQ-INT-04	TC-02, TC-07, TC-08, TC-09
REQ-INT-05 to REQ-INT-07	TC-05, TC-06
REQ-TIME-01 to REQ-TIME-05	TC-04, TC-05, TC-06
REQ-SAFE-01 to REQ-SAFE-07	TC-03, TC-04, TC-05, TC-06, TC-07, TC-09
REQ-EVO-01 to REQ-EVO-05	TC-01, TC-05, TC-06, TC-09, TC-10
REQ-SEC-01 to REQ-SEC-03	TC-01, code/file inspection

7. Appendix A. Implementation Artifact Mapping

Artifact	Purpose in Prototype	Requirement Links
safe_overtake.pla yground	Defines world size, road texture, robot names, initial positions, ports.	REQ-ARCH-06, REQ-SAFE-06, TC-01, TC-10
straight_road.svg	Defines two-lane undivided road, black surface, white boundaries, dashed center line.	REQ-ARCH-03, REQ-INT-01, TC-01, TC-02
A.aesl	Implements slow vehicle A line following, wall detection, LEDs.	REQ-FUNC-01, REQ-FUNC-02, REQ-FUNC-09
B.aesl	Implements fast vehicle B line following, A detection, overtake phases, C danger handling, LEDs/sound.	REQ-FUNC-01 to REQ-FUNC-10, REQ-TIME-01 to REQ-TIME-05
C.aesl	Implements opposite-direction vehicle C line following and wall detection.	REQ-FUNC-01, REQ-FUNC-09
calibrate.aesl	Displays ground and proximity sensor values through LEDs for calibration.	REQ-FUNC-11, REQ-SAFE-04, TC-09

8. Appendix B. Current Implementation Parameters

Parameter / Element	Current Value	Meaning
A baseSpeed	28	A.aesl slow vehicle speed.
B baseSpeed	65	B.aesl faster vehicle speed for overtake scenario.
C baseSpeed	40	C.aesl opposite vehicle speed.
thresholdLeft / thresholdRight	75 / 77	Ground sensor line-following thresholds.
collisionDistance	400	B detects A in front.
dangerDistance	300	B detects C during overtake.
stopDistance	3000	Wall-like obstacle detection threshold.
deadZone	4	Ignores small line-following corrections.
Nominal B phases	1–5	Pull left, align, pass, merge back, straighten.
Emergency B phases	6–8	Clear lane, straighten/resume, reset emergency flags.

9. Appendix C. Glossary

Term	Meaning in this SRS
SoS	System-of-Systems: the complete safe overtake scenario composed of autonomous CSs and the shared environment.

CS	Constituent System: a vehicle or environment element that participates in the SoS.
ECS	Environmental Constituent System: the road/shared physical environment used for indirect coordination.
RUI	Relied Upon Interface: interface of a CS relied on by the SoS to achieve the higher-level service.
RUPI	Relied Upon Physical Interface: sensors/actuators and physical interactions such as ground sensing, proximity sensing, motors, LEDs, road markings.
RUMI	Relied Upon Message Interface: direct cyber-message interface. Not implemented in the current prototype but kept as an extension concept.
Stigmergic channel	Indirect communication through the shared physical environment, such as observing another vehicle position via proximity sensors.
Aseba	Event-based programming environment used for Thymio robot behavior.
Playground	Simulator environment where the road world and Thymio robots are loaded.